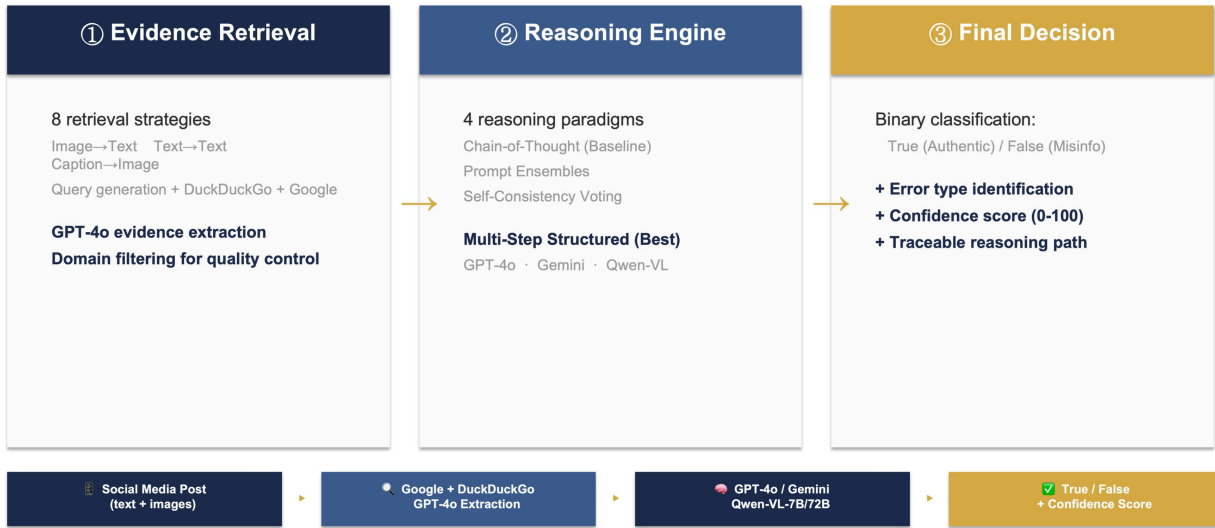


MultiFact

Multimodal Misinformation Detection on Social Media

System Architecture: Evidence → Reasoning → Decision



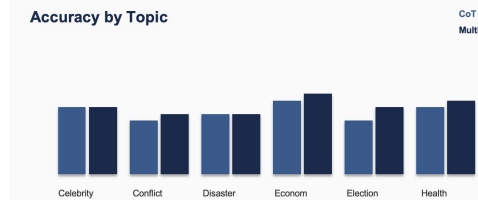
Model Performance: Visual Comparison

Key Takeaways

- Multi-Step: +6.9% accuracy over CoT, +2.8% over Prompt Ensemble — a substantial margin
- Recall 86.1% is the business-critical metric — fewer undetected misinformation posts escape
- Staged verification eliminates image-caption mismatch errors (the dominant CoT failure mode)
- Prompt Ensemble provides marginal gains but cannot fix structural reasoning flaws

CoT Evidence (Baseline)	Prompt Ensemble	Multi-Step Reasoning ★ Best Model
Single-stage Chain-of-Thought: all evidence → one reasoning chain → final label.	Multiple prompt templates voted via majority rule.	Staged verification pipeline:
Simple, interpretable, widely used in misinformation research.	Tests whether prompt diversity stabilizes output and reduces wording sensitivity.	① Caption truthfulness check ② Image misuse detection ③ Cross-modal final verdict
Input: Image-search + Caption-search + Web-text retrieval	Input: Image-search + Caption-search + Web-text retrieval	Input: Image-search + Caption-search → Separates text & visual deception

Accuracy by Topic



Evidence Coverage by Retrieval Source



Error Reduction: Multi-Step vs CoT Baseline

Error Type	CoT	Multi-Step	Change
Ambiguous Context	3	2	Same
Entity/Date Mismatch	5	3	↓ 40%
Image-Caption Mismatch	3	2	↓ 33% — key improvement
Missing Evidence	3	3	Same (shared ceiling)
Satire/Rhetoric	2	1	↓ 50%

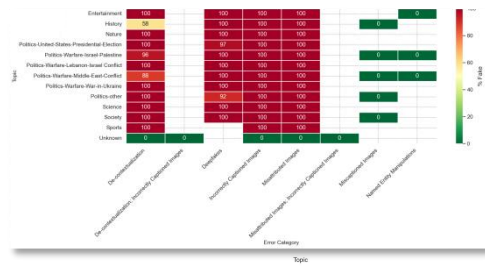
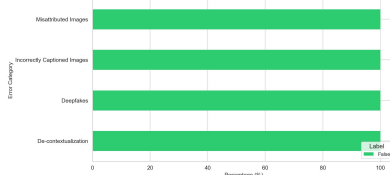


Figure 3.1 — Label Distribution Within Each Error Category



Slice-Based Error Analysis

- Election & Health topics → universal accuracy decline — ambiguous, polarized claims lack definitive evidence
- Image-caption mismatch is the #1 CoT failure → Multi-Step eliminates this via independent image misuse inspection
- Missing evidence affects ALL pipelines equally → retrieval quality is the shared performance ceiling

Ambiguous Context	Entity/Date Mismatch	Image-Caption Mismatch	Missing Reliable Evidence	Satire or Rhetoric
Errors: CoT:3 Ens:2 MS:2	Errors: CoT:5 Ens:4 MS:3	Errors: CoT:3 Ens:2 MS:2	Errors: CoT:3 Ens:3 MS:3	Errors: CoT:2 Ens:2 MS:1
Unclear factual background leads to guesswork	Minor factual errors confused with real misinformation	Real image + fake caption. CoT cannot separate these.	Universal limitation: no evidence → forced to guess	Humorous/satirical tone misread as false claims

Model Limitations & Future Directions

API Dependency

Full pipeline requires paid LLM, image recognition, and search service credentials. Cannot run fully offline.

Retrieval Noise

Search outputs may be outdated, duplicated, or unsubstantiated. Contradictory evidence can distort all models' reasoning.

OOD Generalization

Trained on 2024-2025 English social media posts. Performance drops on long videos, low-resource languages, other content types.

No Pixel-Level Detection

All architectures only evaluate cross-modal semantic alignment. No dedicated module for detecting Photoshop or subtle AI edits.

